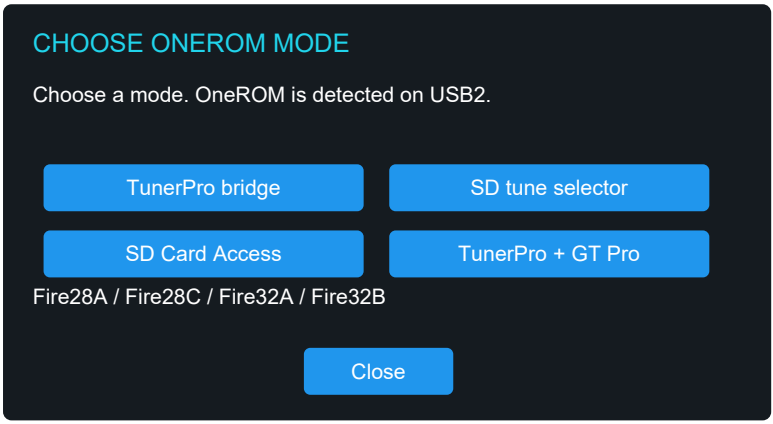
RAPID FIRE SOFTWARE

Rapid Fire Commander Wi-Fi

Wi-Fi handbook

Windows flashing, Wi-Fi setup, OneROM, TunerPro RT, SD tunes and wideband wiring.



OneROM mode chooser in the Wi-Fi build (firmware layout recreation)

RapidFire-001 / rapid-fire-commander-wifi

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Print as a bifold booklet. In a PDF reader with booklet printing, choose **Booklet**, A4 paper, both sides and left binding. If the printer requires a flip setting, choose short-edge duplex. Fold the sheets down the middle.

Wi-Fi setup and bench results

Commander Wi-Fi connects TunerPro RT on a Windows laptop to OneROM through the Guition ESP32-P4 dashboard. The bench build targets the 4.3-inch Guition JC4880P443C_I_W/Y with 16 MB flash and ESP32-P4 silicon revisions 0.1 through 1.99.

The dashboard also includes a FAT32 SD Tune Selector for up to 64 compatible BIN files, USB SD Card Access, and on-screen wideband display. The guides below cover the hardware and menus in more detail.

- Commander menu and feature guide covers OneROM, SD tunes, settings and GT Pro logging, with screenshots and a Wi-Fi connection diagram.

- Windows flashing and recovery covers board identification, BOOT mode, backups and using the private Wi-Fi kit.

- Wideband wiring and setup covers the Expand IO pinout and the Innovate LM-2 and AEM options.

The Wi-Fi firmware and Windows kit are distributed privately. This repository contains the documentation and printable booklet.

Connect the laptop

1. With the Wi-Fi build already installed, connect OneROM to the P4's **USB2** port and power the P4 through **USB3**.

2. On the P4, open **OneROM > TunerPro bridge** for emulation, or **OneROM > TunerPro + GT Pro** to include Holden ALDL logging. Wait for **OneROM ready on USB2**.

3. In Windows, join the network shown on the P4's **Commander Wi-Fi** screen using the password displayed there.

4. Open the Commander Wi-Fi Windows app and select **Connect**.

5. In TunerPro RT, select the COM port mapped to the OneROM endpoint. For logging, select the GT Pro under **Settings > OBDX devices** on the P4, choose the separate acquisition COM port in TunerPro RT, and load the ADX that matches the ECU and calibration.

TunerPro RT function	Local endpoint for the Windows COM mapping
OneROM emulation	`127.0.0.1:22170`
GT Pro ALDL acquisition	`127.0.0.1:22171`

The Windows app presents these as local RFC 2217 endpoints. Use the COM numbers assigned by your Windows serial port provider in TunerPro RT. The Commander Wi-Fi app has no trial timer.

Bench results

On the owner's Windows laptop, Commander connected over Wi-Fi and TunerPro RT detected OneROM. The owner reported a successful BIN download and upload with no errors. The matching ADX loaded in TunerPro RT. Live GT Pro telemetry with an ECU connected remains to be verified on the Wi-Fi build.

Troubleshooting

- **Peer disconnected** can occur when the P4 is waiting for OneROM. Check that OneROM is on **USB2** and wait for **OneROM ready on USB2** before selecting **Connect** in the app.

- If a fresh session is needed, close TunerPro RT, disconnect the Windows app and restart the P4. Select the bridge mode again and reconnect the laptop to the dashboard's Wi-Fi.

- For the wired release and its printable handbook, see the Commander README.

Copyright

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Flash on Windows

Check your board

This kit targets the **Guition JC4880P443C_I_W/Y**, a 4.3-inch, 480 × 800 touch display with an ESP32-P4, ST7701S LCD controller and 16 MB flash. The Commander build is for the early ESP32-P4 silicon range accepted by the image, v0.1 through v1.99. Use the exact board and verify its chip revision. Similar-looking Guition displays can have different panels, flash layouts or ports. Guition board specifications

Port	Use
USB2	Windows flashing in bootloader mode. In Commander, this port hosts the OneROM.
USB3	Power during Wi-Fi operation or connect Windows for SD Card Access.

Use a data-capable USB cable and Windows x64. Disconnect OneROM and any vehicle wiring for flashing. The merged image replaces the full 16 MB of P4 flash, including saved settings. Back up the original image first if you want a way back. This kit updates the P4 and leaves the separate wireless coprocessor firmware alone.

Enter the bootloader

1. With the private Commander Wi-Fi kit available, use **Extract All** on its ZIP. Keep the launcher, firmware and **tools** folder together.

2. Open **Device Manager > Ports (COM & LPT)** so you can see which port appears.

3. Unplug the USB and external power cables from the P4. Hold the board's **BOOT** button while reconnecting a Windows data cable to **USB2**. On the owner's tested unit this was the leftmost button. Use the board marking or vendor instructions on other revisions.

4. Release BOOT after Windows shows a new COM port. The LCD can stay blank in the bootloader. Write down the COM number. If several ports appear, compare the list with the board unplugged and re-enter bootloader mode.

Optional backup

Open PowerShell in the extracted folder. Replace `COM7` with your actual bootloader port. Pick a new filename and store the backup privately. Keep the P4 connected until the command completes.

```
.\tools\esptool.exe --chip esp32p4 --port COM7 --baud 115200 `
  --before no_reset --after no_reset `
  read_flash 0x0 0x1000000 p4-original-backup.bin
```

Flash and check the result

1. Double-click **Flash Commander.cmd**. Enter the P4's COM port. The launcher checks the firmware checksum and asks esptool to identify the chip, revision and flash size before offering to write.

2. Review the details. Type **FLASH** only for the intended board. The launcher writes the **merged** image at **0x0** and esptool checks the data it wrote. Keep the cable attached until the result is shown.

3. After success, release BOOT if you are still holding it. If the P4 stays in the bootloader, unplug and reconnect its power with BOOT released.

4. Look for the **Rapid Fire Commander** fade-in followed by the **Commander** dashboard. Verify touch response and check **Settings > Brightness**. Reconnect OneROM after the first boot check.

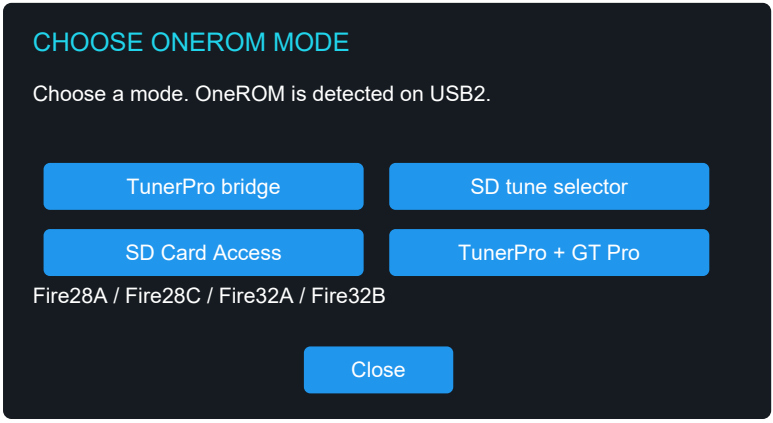
Troubleshooting

Symptom	What to check
No COM port	Use a known data cable directly to the computer. Check USB2 and repeat the BOOT sequence. Look in Device Manager for the new device.
Failed to connect or port busy	Close TunerPro, serial monitors and other flash tools. Confirm the COM number and repeat the BOOT sequence.
The launcher rejects the chip, revision or flash size	Compare the board label and chip identification with the compatibility section above.
Flash verification fails	Keep OneROM unplugged. Check cable and power, re-enter bootloader mode and retry the same kit.
Screen stays blank after success	Boot normally with BOOT released. If it stays blank, re-enter the bootloader, check the exact board model, and use the original backup for recovery if needed.
COM port disappears during normal boot	USB2 changes roles in Commander. Select **TunerPro bridge**, connect the Windows app over Wi-Fi, then use the mapped COM port in TunerPro RT.

The kit uses the official Windows x64 esptool v4.9.1 release.

Commander menus and tuning

The shared menu illustrations come from the wired Commander documentation. Status values and filenames are examples. The connection diagram below shows the Wi-Fi bridge.



OneROM mode chooser in the Wi-Fi build (firmware layout recreation)

Program and verify OneROM in the Rapid Fire OneROM app

1. Keep OneROM disconnected from the ECU. Connect it directly to your computer and open the Rapid Fire OneROM app.

2. The Rapid Fire OneROM app automatically detects and selects your OneROM board and sets the intended ROM size. Select the **NORMAL DT** firmware profile, then program the tune BIN you want to start with.

3. Read the programmed ROM back in the app and confirm that it matches the intended BIN byte for byte. If you plan to use the SD Tune Selector, put a copy of that verified BIN at the root of the FAT32 microSD card.

4. Once the readback matches, disconnect OneROM from the computer and plug it into Commander's **USB2** port. Open the Commander mode you want to use and confirm that it detects the expected board and ROM size.

OneROM modes

Mode	Use
TunerPro bridge	Laptop on Commander Wi-Fi, OneROM on USB2. Select the Windows COM port mapped to OneROM emulation.
TunerPro + GT Pro	Separate Windows COM ports for OneROM emulation and read-only GT Pro ADX logging.
SD tune selector	Browse and load a matching BIN from the FAT32 microSD card onto OneROM with the engine off.
SD Card Access	Expose the P4's SD card as a removable drive through USB3. The tune selector and serial bridge pause while this mode owns the card.

TunerPro bridge

Connect OneROM to **USB2** and power the P4 through **USB3**. Open **OneROM > TunerPro bridge**, wait for **OneROM ready on USB2**, join the P4's Wi-Fi in Windows and select **Connect** in the Commander Wi-Fi app. Choose the COM port mapped to OneROM emulation in TunerPro RT. Start with a readback before attempting a write. See COM port mapping.

With a compatible OneROM setup and emulation enabled, TunerPro RT can send live calibration edits while the engine is running. TunerPro RT sends the editor update through Commander's bridge, and OneROM serves the updated data to the ECU. Perform live edits while the vehicle is stationary. See the TunerPro RT emulation guide.

Wi-Fi connection paths

Connection	Path
Laptop to P4	TunerPro RT > Windows COM ports > Commander app > Wi-Fi > P4
OneROM emulation	P4 > USB2 > OneROM
Read-only GT Pro logging	P4 > Bluetooth > GT Pro > ECU via ALDL

TunerPro + GT Pro ADX logging

Commander Wi-Fi passes read-only ALDL requests and replies between TunerPro RT and the GT Pro. Load the ADX in TunerPro RT so it can decode the ECU replies.

1. Connect OneROM to **USB2** and power the P4 through **USB3**. Choose **OneROM > TunerPro + GT Pro**, join the P4 network in Windows and connect the Commander Wi-Fi app.

2. In **Settings > OBDX devices**, select the GT Pro and tap **Connect selected**. Wait for **GT Pro ALDL acquisition ready** on the OneROM screen.

3. In TunerPro RT, use the **OneROM emulation** COM port for Ostrich. Under **Preferences > Data Acq./Emulation**, choose **Standard Serial** and the separate **GT Pro ALDL acquisition** COM port. Use the COM numbers assigned by your Windows serial port provider.

4. Stop acquisition if it is running. Choose **Acquisition > Load Definition File**, load the ADX for your ECU and calibration, then start acquisition.

After a verified SD tune load, Commander can identify OSE $11P V104 or $12P V112. Use that recommendation to choose the matching ADX variant in TunerPro RT. For a factory or unknown profile, choose the definition for the installed ECU and calibration.

See Wi-Fi bench results for the confirmed tests. Live GT Pro telemetry with an ECU connected remains to be verified on the Wi-Fi build.

SD Tune Selector

The SD Tune Selector loads the selected BIN image in blocks. Even when two BINs share the same ECU program code, an ECU reading during the transfer can see a mix of old and new calibration data. For SD tune swaps and full BIN uploads, OneROM can stay installed in the vehicle's MemCal. Keep the engine off throughout the upload and verify the readback before starting it.

Fire model	Supported ROM size
Fire28A, Fire28C	16, 32 or 64 KiB
Fire32A, Fire32B	128 KiB, 27C010

Commander uses model and active-size metadata from OneROM. Use the matching OneROM v0.8.0 **NORMAL DT** firmware. Unsupported or unreadable profiles stop before a load.

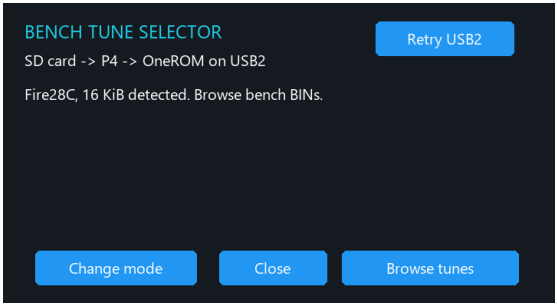
1. Copy the candidate BINs and a verified copy of the current BIN to the root of a FAT32 card. Use descriptive filenames.

2. Choose **SD tune selector** and check the detected board and ROM size.

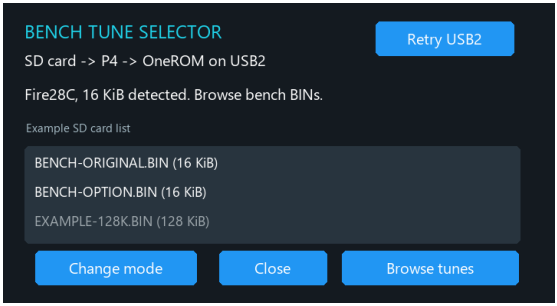
3. Tap **Browse tunes**. The list shows up to 64 BIN filenames and sizes. Files of a different size are disabled.

4. Select a compatible file and confirm the load. Commander checks the existing image against the card and validates the expected ROM layout before writing. For 32, 64 and 128 KiB images, identical halves fail this check.

5. Wait for **Verified byte for byte**. If a write is incomplete or verification fails, keep the engine off and restore and verify the original BIN before starting it using your OneROM recovery procedure.



Fire28C SD Tune Selector ready to browse

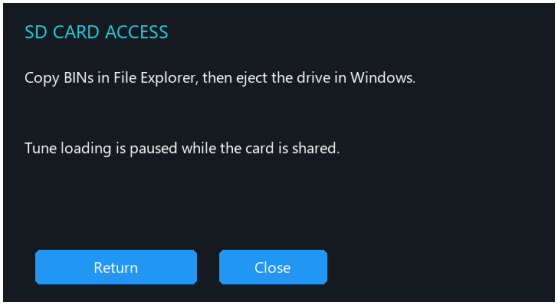


Example SD BIN list with a disabled size mismatch

Tap **Retry USB2** if the selector loses the device. The P4 can restart its USB2 host and the OneROM session while the cable stays in place. A retry does not replay a write. For **Change mode**, finish transfers, close TunerPro, turn the engine off and confirm the restart. A restart can discard unsaved tune changes.

SD Card Access

Choose **SD Card Access** and connect Windows to USB3. Copy files to the card in File Explorer. Eject the drive in Windows and wait for the P4 to show that the card was released. Tap **Return** and select the next mode after the restart. You can also copy files with a separate card reader.



SD Card Access with Windows eject instructions

Dashboard, Engine and Settings

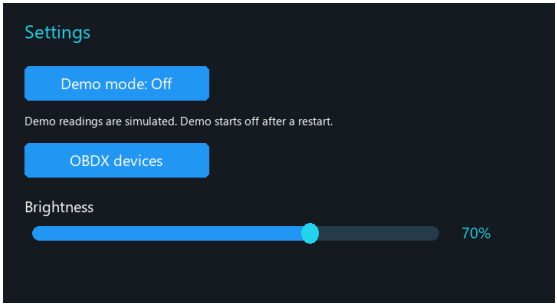
- Dashboard and **Engine** show speed, RPM, intake air, coolant, spark advance and battery fields. Unavailable readings display dashes.

- Settings > Brightness adjusts the backlight from 10% to 100% and saves on release.

- Settings > Demo mode supplies simulated gauge values, marked by a **DEMO** badge. It starts off after a restart.

- Settings > OBDX devices opens Bluetooth discovery, connection controls and diagnostics. Use **TunerPro + GT Pro** to log ECU data in TunerPro RT.

- Wideband > Source chooses Innovate LM-2 or AEM 30-0300. See wideband setup for wiring and signal levels. Use one source at a time.



Settings with Demo mode and brightness



Rapid Fire Commander startup logo fade

Wideband wiring

Commander receives a digital wideband signal on **P4 UART2 RX, GPIO32**. Use one supported source at a time. The screen shows AFR and lambda when valid data arrives.

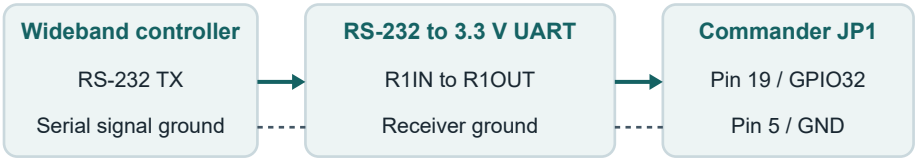
Find the wideband input on the display board

On the Guition **JC4880P443C_I_W** schematic, **JP1** is the 2×13 header labeled **Expand IO**. Look for **GPIO32** at JP1 pin 19 and **GND** at pin 5. The schematic shows pin 20 beside pin 19 as **C6_U0RXD**. That pin belongs to the ESP32-C6. The separate RS-485 connector uses different signals. Match the board model and pin-1 orientation to the drawing before wiring. Guition Expand IO schematic

Required serial interface

Use a **true RS-232 to 3.3 V UART receiver**, for example a MAX3232 breakout designed for 3.3 V operation and powered at 3.3 V. Check the breakout pinout for its RS-232 receive input and its 3.3 V logic receive output. Keep raw RS-232 voltages, 5 V logic, vehicle power, CAN and analog outputs away from the P4 GPIO. A USB-to-TTL cable alone does not convert RS-232 signal levels. MAX3232 reference

Wideband controller wiring



Solid arrows show data. Dashed lines show ground. Leave RS-485 unused.

From	To
Wideband RS-232 TX	Receiver RS-232 input, often labeled R1IN
Receiver 3.3 V logic output, often labeled R1OUT	Guition JP1 pin 19, P4 GPIO32
Receiver logic ground	Guition JP1 pin 5, GND

From	To
Wideband serial signal ground	Receiver RS-232 signal ground, according to the controller and receiver pinouts

The P4 connection is receive-only. Leave adapter transmit wiring to the wideband unused. Power the module from a suitable regulated 3.3 V source as specified by the module. Check the JP1 orientation and actual board revision with power off. Guition board specifications

Choose the source

Controller	Signal	Commander setting
Innovate LM-2	MTS Serial OUT. Use the matching MTS serial-out lead for the LM-2 connector and verify its TX and ground pinout.	**Wideband > Source: LM-2**. The firmware listens at 19,200 baud, 8N1 and uses the first channel.
AEM X-Series 30-0300	Blue RS-232 Serial OUT wire. Set the gauge to AFR display mode.	**Wideband > Source: AEM**. The firmware listens at 9,600 baud, 8N1 and derives lambda using a gasoline reference of 14.65.

Compare the P4 reading to the controller display on the bench. Dashes indicate unavailable input or invalid readings. The dashboard returns to LM-2 after a reboot. This feature displays readings on the P4. TunerPro wideband passthrough remains pending, and both sources still need physical verification on this build. Innovate LM-2 manual · AEM 30-0300 manual

Windows app walkthrough

Extract the private Commander Wi-Fi Windows kit before opening the app.

Start a session

1. In Windows Wi-Fi settings, join the network shown on the P4 Commander Wi-Fi screen. Enter the password displayed there.

2. Open **Rapid Fire Commander Wi-Fi.exe** and select **Connect**. Wait for **Connected**.

3. Leave the app open. Select the mapped OneROM COM port in TunerPro RT. For GT Pro logging, select the acquisition COM port and load the matching ADX.

Stop or restart a session

Close TunerPro RT before choosing **Disconnect** in Commander. To start fresh, restart the P4, select the bridge mode, rejoin its network and select **Connect** again.

Rapid Fire Software

For setup or recovery help, include your board model, the Commander build, the mode selected and the exact error message.

support@rapidfiresoftware.net

Commander Wi-Fi guides on GitHub